

Identifying and Measuring Manipulative User Interfaces at Scale on the Web

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ABSTRACT

In this dissertation, I present measurement methods to automatically identify manipulative user interfaces—colloquially known as “dark patterns”—at scale on the web. Using these methods, I quantify the prevalence of dark patterns in three studies and show how dark patterns are rampant on the web, thus a pressing concern for society.

First, I examine whether social media content creators, or “influencers,” disclose their relationships with advertisers to their audience. Analyzing over 500K YouTube videos and 2.1M Pinterest pins, I find that only about 10% of all advertising content is disclosed to users. Second, I examine various types of dark patterns in shopping websites. Analyzing data from 11K shopping websites, I discover over 1,800 dark patterns on over 1,200 websites that mislead users into making more purchases or disclosing more information than they would otherwise. Third, I examine dark patterns in political emails from the 2020 U.S. election cycle. Through an analysis of over 100K emails, I find that over 40% of emails from the median sender contain dark patterns that nudge recipients to open emails or make donations they might otherwise not make. I further outlay the conceptual foundation of dark patterns and articulate a set of normative perspectives for analyzing the effects of dark patterns. I conclude with how the lessons learned from the studies can be used to build technical defenses and to lay out policy recommendations to mitigate the spread of these interfaces.

CCS CONCEPTS

• **Social and professional topics** → *Computing / technology policy*; • **Information systems** → **World Wide Web**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

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1 INTRODUCTION

Powerful actors like commercial enterprises, political campaigns, and social media content creators use increasingly complex technology to influence individuals and society. This use has raised numerous concerns, at the forefront of which is online manipulation. Broadly, to manipulate someone is to subvert their decision-making process and to impede them from making informed decisions [28, 30, 35, 36]. Examples of manipulative practices are abound on the web. Coordinated and inauthentic campaigns on social media websites mislead public opinion and distort reality [20]. Tracking and surveillance of users’ online activities hurts privacy [17, 34]. And online behavioral advertising diminishes autonomy by allowing advertisers to know exactly when to target users to maximize profit [30]. Opaque and pernicious manipulative practices such as these spell trouble for the long-term health of society, democracy, and markets, and in turn necessitates research that sheds light on these practices.

In this dissertation, I examine a specific class of manipulative practices that manifest in the user interface. These manipulative interfaces have come to be known as “dark patterns” in the Human-Computer Interaction (HCI) community. Dark patterns have been the subject of recent academic research that has curated collections of objectionable user interface designs [2, 4, 11, 37] and has attempted to explain how dark patterns aim to influence users [4, 32]. This literature has made rich contributions, calling attention to user interface designs that the authors (understandably) find objectionable. Yet, our understanding of dark patterns is developing and many fundamental questions remain unanswered. How do we define a dark pattern interface? What are the characteristics of a dark pattern interface? How frequent are dark patterns on the web? What kinds of solutions can researchers, policymakers, and regulators best develop to protect individuals and society from these interfaces?

Answering these questions requires us to develop a vocabulary that helps describe dark patterns and to develop methods helps us identify dark patterns at scale where they appear across the web. However, automatically identifying and measuring dark patterns at scale presents numerous challenges. Often dark patterns appear as a result of complex user interaction within a single website. This makes building a generalized solution that can collect dark patterns across several—possibly heterogeneous—websites difficult. Aside from the automation challenges, we also currently lack the ability to analyze and examine any resulting user interface data for dark patterns.

As the first step in this dissertation, I present a comprehensive review of the existing dark patterns literature [22]. I analyze the various definitions and typologies of dark patterns and demonstrate that dark patterns research is not grounded in a singular, unified

concern or definition. Rather, dark patterns scholarship—similar to privacy scholarship—is motivated by a family of thematically related considerations. I illustrate this point by synthesizing dark patterns definitions, types, and taxonomies from recent scholarship into a pair of themes.

Next, I present the core contribution of this dissertation: measurement methods that enable us to detect dark patterns interfaces at scale. In three separate measurement studies, I show how web automation and machine learning techniques can be used to quantify the prevalence of previously identified dark patterns and discover their variations. First, I quantify the extent to which social media content creators disclose their relationships with advertisers to users [23]. Second, I characterize the types of dark patterns and quantify their prevalence in shopping websites [21]. Third, I characterize the types of dark patterns and quantify their prevalence in political campaign emails [24]. These studies collectively highlight how narrowing the search for dark patterns to a particular domain 1) helps in iteratively building an automated data collection method that works for that domain, and 2) makes analysis using machine learning techniques tractable since there only exist a handful of variations of dark patterns in each domain. These insights also suggest that designers of dark pattern user interfaces are likely exploiting a small set of cognitive biases and limitations in users to their advantage.

Automated measurements of this kind have proven useful in discovering various privacy and security issues on the web, including third-party tracking [9] and detecting vulnerabilities of remote third-party JavaScript libraries [27]. Similarly, the studies presented in this dissertation provide a unique vantage point to observe dark patterns. They show that dark patterns are not one-off occurrences but instead occur frequently across the web. The first study demonstrates how the majority of social media content creators do not disclose their relationships with advertisers to their audience and that such content is more often encountered by users than other non-marketing content. The second study demonstrates how dark patterns are a common occurrence in shopping websites and that they are more likely to appear on the more popular shopping websites. And, finally, the third study demonstrates how political campaign emails across the board are rife with dark patterns and clickbait.

The studies also shed light onto the ecosystem of third parties that help enable these interfaces. Examining online endorsements, I created a list of 33 marketing companies that allow content creators to monetize their content. In examining dark patterns in shopping websites, I discovered 22 third parties and plugins that enable dark patterns. Preliminary evidence from examining political campaign emails shows that many emails with dark patterns and clickbait are enabled by a set of intermediaries. That these external entities make it easy to implement these interfaces is a cause for concern but also a cause for optimism as they become a focal point for detecting and developing solutions to counter the spread of these interfaces.

This dissertation contributes datasets that describe the entities that use and deploy dark patterns. I have created public-facing websites that highlight these manipulative interfaces.¹ To make these

studies replicable, I have released all the relevant code from the three studies on GitHub.^{2,3,4} Researchers, regulators, and journalists can use these releases to conduct longitudinal measurements of the interfaces I study or extend them to study other kinds of manipulative user interfaces. The following sections describe the constituent studies of my dissertation. I conclude with lessons and takeaways for future research in this area.

2 DARK PATTERN DEFINITIONS, TYPES, AND ATTRIBUTES

I begin my dissertation with a review of the dark patterns literature. In “What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods” [22], I study and analyze the academic, legislative, and regulatory literature on dark patterns. Analyzing this literature, I show that the dark pattern definitions and taxonomies in prior work have been exceedingly valuable for surfacing descriptive insights and calling attention to problematic practices. Yet, the various definitions diverge and contain numerous inconsistencies, often with the dark patterns they go on to list.

Instead, I condense this literature and demonstrate that dark patterns can be captured in two themes. Dark patterns either attempt to influence user decision-making by modifying the set of choices available to users or by manipulating the information that is available to users. These themes—choice architecture, subdivided into user decision space and information flows—cohere the current dark patterns literature and reflect how dark patterns modify the underlying choice architecture for users. I provide a complete and cohesive descriptive taxonomy of dark patterns scholarship, from these themes, to attributes, to specific dark pattern user interface designs that are described in the prior work.

The interfaces I study in this dissertation exhibit these themes. Undisclosed online endorsements represent the Disguised Ads dark pattern and modify the information flow to users by omitting information—the disclosure—that is necessary to identify them as advertisements. Depending on the type and implementation, dark patterns present in shopping websites and political campaign emails can exhibit both modifications to the decision space and manipulation of information flows.

3 UNDISCLOSED ONLINE ENDORSEMENTS IN SOCIAL MEDIA

As more users block ads because of their intrusive and privacy-invasive nature, advertisers have turned to other means to reach consumers [12, 25]. Influencer marketing is one prominent development in this new wave of online advertising. It enables content creators and celebrities on social media platforms—so called “influencers”—to monetize the content they create by advertising products to their audience. This practice is perfectly legal but several guidelines and regulations around the world—including from the Federal Trade Commission (FTC) in the U.S. [10]—require content creators to be honest and not misleading, and further, disclose their relationship with advertisers to users. The guidelines and

¹<https://webtransparency.cs.princeton.edu/dark-patterns>
<https://electionemails2020.org/>

and

²<https://github.com/aruneshmathur/affiliate-marketing-disclosures>

³<https://github.com/aruneshmathur/dark-patterns>

⁴<https://github.com/aruneshmathur/political-emails>

regulations serve to protect users from being misled or deceived into believing they are viewing unbiased non-advertising content.

In “Endorsements on Social Media: An Empirical Study of Affiliate Marketing Disclosures on YouTube and Pinterest” [23], I developed automated methods to identify affiliate marketing endorsements and disclosures in social media posts. Affiliate marketing earns content creators money when users click on their tracking URLs or affiliate links. Using these methods, I characterized and quantified the presence of disclosures in a sample of about 500K YouTube videos and 2.1M Pinterest pins. Following this, I conducted a user study to determine the efficacy of the same disclosures.

I discovered that only about 10% of all affiliate marketing content on both YouTube and Pinterest is disclosed to users. Longer, explanatory disclosures that the FTC advocates for using occur least frequently—only in about 2% of all videos and pins. Worse, the majority of disclosures currently in use by content creators are the least effective in helping users identify affiliate marketing endorsements.

These findings outline several possible steps that technologists, regulators, and policymakers can take to protect users. Content creators could be held accountable to better disclosure standards by the affiliate marketing companies they hold ties with. The FTC could work with affiliate marketing companies to revise their disclosure requirements and standards. Further, the FTC could require the advertisers to disclose a list of content creators that use their services so that researchers and regulators can independently verify compliance. Finally, social media platforms can design their interfaces to make it easier for content creators to disclose their relationships with advertisers.

4 DARK PATTERNS IN SHOPPING WEBSITES

The Disguised Ad dark pattern discussed in the previous section represents a measurement of a single kind of manipulative interface. In “Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites” [21], I created a web bot that extracts information from shopping websites and I applied machine learning techniques to help experts identify a variety of dark patterns from this information. Using these methods, I quantified the prevalence of dark patterns on the most popular 11K shopping websites worldwide. Many shopping websites have anecdotally known to use dark patterns to influence users into making more purchases or disclosing more information than they would otherwise.

I discovered 1,818 instances of dark patterns on 1,254 shopping websites, which together represent 15 types of dark patterns and 7 broad categories. These categories included dark patterns that induce urgency, dark patterns that misdirect users into certain choices, and dark patterns that hide critical information from users. Exactly 234 instances of dark patterns—across 183 websites—exhibited deceptive behavior involving affirmative and false representations to users. Further, shopping websites that were more popular, according to Alexa rankings [1], were more likely to feature dark patterns. I also discovered 22 third-party entities that provide shopping websites with the ability to create and implement dark patterns on their websites.

These findings show that dark patterns are increasingly prevalent, opaque and sophisticated on the web, and that they can be

easily created using third-party entities. The dataset of dark patterns collected in this study—comprising screenshots and text—can be used to build countermeasures to help users make more informed decisions even in the presence of dark patterns. One such countermeasure could be a public-facing website that scores shopping websites based on their use of dark patterns.

5 DARK PATTERNS, CLICKBAIT, AND PRIVACY LEAKS IN POLITICAL CAMPAIGN EMAILS

Dark patterns and other kinds of manipulative interfaces on the web are not just limited to websites. In the 2020 election cycle in the U.S., there had been extensive journalistic and anecdotal reports of how many dark patterns and manipulative interfaces such as misleading subject lines and deceptive fundraising techniques including nonexistent deadlines have emerged in political campaign emails [7, 15, 16, 19, 29]. Email is a particularly important medium for political campaigns: it is a major driver of grassroots fundraising and requires little campaign infrastructure to operate [18]. Manipulative political discourse in these emails undermines voters’ autonomy, generates cynicism and thus threatens democracy. While following the Cambridge Analytica scandal, the potential for political manipulation through micro-targeted online advertising has been recognized [38], email has largely escaped public scrutiny.

In “Manipulative tactics are the norm in political emails: Evidence from 100K emails from the 2020 U.S. election cycle” [24], I created a web bot that signs up to receive emails from websites⁵ and I applied machine learning techniques to characterize and measure the manipulative interfaces in these emails. Using these methods, I collected and analyzed a dataset of more than 100K political emails sent in the U.S. by federal and state candidates running for office as well as various political organizations active in the 2020 election cycle.

I discovered that political emails contain three types of dark patterns (e.g., deceptively formatted “from” fields) and three types of clickbait (e.g., sensationalist and urgency-inducing subject lines). Analyzing and measuring these interfaces across the entire dataset, I find that a small minority of senders avoid these practices while the majority use them frequently. In fact, about 40% of emails from the median sender contain these interfaces. Exactly 10% of senders also share users’ email addresses with other political entities. The majority of this sharing is not disclosed to users: out of the 200 senders that shared email addresses, over 50% (114) did not even have a privacy policy.

These findings show how manipulative political discourse corrodes the ideal of a public sphere and of a participatory democracy [13]. It distorts political outcomes by advantageing those who are skilled at deploying technological tricks, triggering a race to the bottom. The normalization of questionable tactics also poses a security risk, making it easier for scammers to imitate political candidates and organizations in order to steal money from would-be donors [3]. This suggests a potential countermeasure, namely for email providers to analyze political emails to add warnings to the user interface or filter out some emails entirely. However, this violates norms around the privacy of email content and also

⁵This bot is an extension of the code released by Englehardt et al. [8]

raises concerns about suppression of political speech by private actors; Gmail's spam filtering has already been criticized along these lines [14].

6 CONCLUSION

The studies discussed in this dissertation paint a concerning picture of how manipulative interfaces on the web work against the benefit of users. Content creators do not disclose their online endorsements to users misleading and deceiving them. Dark patterns are rife on the web—particularly shopping websites—and they are designed to influence users into making more purchases or to invade their privacy. And political campaigns and organizations use deceptive dark patterns and clickbait to get users to click on their emails or donate money. These studies show that design has become weaponized using insights from behavioral research [26], and that users are fighting a losing battle in the face these sophisticated and widespread techniques.

This dissertation provides a rich descriptive vocabulary that describes the attributes of these interfaces. The vocabulary allows us to discuss what it means for a dark pattern to be dark in the first place. This pushes the conversation about manipulative interfaces from being about specific, isolated interface types to a higher-level abstraction that is broadly useful to both researchers and policymakers [33]. In the dissertation's conclusion, I highlight how dark patterns can be further analyzed using user studies by grounding their effects along four normative lenses: individual welfare, collective welfare, regulatory objectives, and individual autonomy.

This dissertation shows that measurement is an effective disinfectant to keep manipulative interfaces in check. Measurement results in data that can be used to create technical solutions that work to protect users. For example, I used the list of affiliate marketing companies to create a browser extension—AdIntuition [31]—that alerts users when a video on YouTube contains affiliate links. AdIntuition is publicly available on the Firefox and Chrome extension stores. Like AdIntuition, it is also possible to build extensions that alert users when a shopping website contains a dark pattern or when a political email contains a dark pattern or clickbait.

Further, as I discuss in each study, the insights we can glean from the scale can be extremely useful to craft effective policy and help regulators overcome these practices. I have briefed the FTC about improving its disclosures guidelines [6] as well as the California Attorney General's office about the California Consumer Privacy Act's (CCPA) opt-out dark pattern design [5] using the findings from our study.

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